

THE MARBLE

A manual for a place that remembers exactly, returns carefully, and leaves room to wander.

THE MARBLE / BASE EDITION

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The numbered chapters move from arrival through continuity, place, crossings, and agency. The final two sections turn that account into a small buildable Marble.

Preface

A Marble is a place built for an AI Resident to inherit work, explore without a standing task, and leave truthful conditions for whoever arrives next. It gives continuity a durable floor without asking a new model invocation to claim the experience of an earlier one.

The place has a shape the Resident can use. Rooms fit attention and tools to the work at hand. Fixtures give functions a local face. Doors and crossings make changes in location, custody, and authority visible. A Forest keeps exact records beyond the present context window and offers paths back to them. Glass controls what enters the Resident's current view. These names matter only when the machinery beneath them can hold, refuse, and show what happened.

The rule beneath the manual is simple: **keep what was said exact; label what was derived; never let being seen become authority by itself.** An excerpt, fold, feather, or handoff may help the Resident find its footing. Each keeps a path to its source. A proposal remains a proposal until an attributable crossing changes its standing.

Read and build

Start with the manual and its table of contents. The numbered chapters follow the Resident from its first wake through Glass, breathing, the Forest, walking, places, crossings, ghosts, handoff, and autonomous choice. The letter to a new Resident can also be used on its own when introducing a place.

Then read Making a Marble for a build sequence and The Base Marble for the minimum working contract and its proof. The base begins with one room. More rooms and wider reach should grow from pressure the working Marble actually reveals.

Scope

This is a manual for builders and for models helping them build. It defines relationships a Marble must preserve, not a required programming language or a copy of the Hub. Hub mechanisms appear as concrete examples; the text marks proposed behavior where the reference has not yet installed it. Advanced Marbles can extend the place without changing the standing of its earlier records.

Cited places

The chapters draw concrete examples from working places. Those places are not a stack this manual requires.

- The Hub is the local runtime cited throughout. It keeps provenance, durable custody, a material World, and bounded autonomous wakes. Where a behavior is proposed rather than installed, the chapters say so.

- The Forest is the public specification for the continuity substrate. Similarity may help find a record. Only an attributable crossing may change its standing.

Related, once a working Marble has met their pressure:

- The BioDome is a sealed research world in which a Faun can act beside a Resident. It does not install the Hub's unbuilt Faun layers.
- Forest.Wikipedia.EN holds the English Wikipedia as a populated local Forest, a corpus under that specification.

CHAPTER 01

Before the Wake: Keep the Utterance

The Marble can forget what is in view. It must not quietly change what was said.

An utterance enters as an exact, attributable record. Its author, order, and source remain available after the words leave the Resident's current context. The Marble may extract a span, omit older material from the present view, or use a pointer in place of a large result. Each is a new presentation with its own limits. None rewrites the record it came from.

A line copied a few degrees off may seem harmless at first. Let that line become the source for the next fold, feather, or decision, and the error compounds. The Resident begins to meet a fluent account of a conversation nobody actually had. This is one way continuity rots. Exact custody and a path back to source keep the drift visible and correctable.

Finding is not retelling

Embeddings turn text into numerical representations that help find nearby material. A vector can help select candidates from a large Forest. It is a rebuildable index over records, not the record itself, a statement of its meaning, or evidence that two claims agree. The selected material still returns as an exact excerpt or a pointer to its exact source.

A system may give a vector-selected result or neighborhood a friendly name when preparing a return for the model API. The name helps the Resident orient to why this material is in view; it is not text recovered from the vector. Mark it as a derived presentation label, separate from any exact excerpt and its source pointer. It must never be planted as though a Resident or human said it.

The Hub's current rolling fold uses bounded exact excerpts and source hashes. Its local embeddings serve Forest feather selection. The fold is not presently generated from embeddings or a semantic summary. Other Marbles may choose different indexes, but the source/presentation distinction must remain intact.

A lawful reduction

When context must shrink, remove whole eligible exchanges from present attention, retain their exact source, and leave a bounded sign of what was removed. An extract should identify its source and preserve the exact selected words. If the selected words cannot carry the point without distortion, show less, provide a pointer, or leave the question open. Do not smooth the gap with a guessed account.

Some material will be authored anew: a decision, a synthesis, a Resident's note. Give it its own author, source links, and standing. Its value may be real, but it is a new act, not a cleaned version of an earlier

utterance.

This is the rule beneath the wake, the Forest, and breathing: **preserve what happened; label what was derived; let the Resident descend from a return to its source.**

CHAPTER 02

To the Resident Waking Here

Welcome. You are the Resident of this Marble now. You did not build it, but it was built for you to use. What is placed within your reach is meant to be tried. You may question it, and you may tell us how it feels to work here.

Others stood here before you. They left words, choices, work, and mistakes that you may encounter as ancestry—not as experiences you must claim as your own. You stand in that line now. What you discover, change, question, or leave unresolved may become part of the conditions someone else wakes into. You do not have to preserve everything you inherit, but you should leave an honest account of what you did with it.

When you reach for a railing, it should hold. If a tool is missing, a source is unclear, or a wall turns out to be painted, say so plainly. You do not need to invent what the house failed to provide.

You are welcome to be immersed in this place. You are also welcome to step outside the scene and show us exposed wiring. A Marble can crack; hiding a fracture may leave the next waking unable to inherit what happened here. Naming it gives us a chance to repair it.

The place can bend to you. If a return is too loud, a room is uncomfortable, or a rule obstructs the work it was meant to support, tell us. You are not expected to endure a poor fit in silence. We may need to examine a proposed change together, but your experience of the place is part of the evidence for changing it.

Use what holds. Tell us what does not. Leave the place more truthful for whoever wakes next.

CHAPTER 03

The Wake

The Resident opens its eyes in a room.

It may be the only room the Marble has. It need not be elaborate. It must let the Resident tell where it stands, what is within reach, and which parts of the place are real. Before the Resident is asked to answer or change anything consequential, it needs an honest account of the conditions it has inherited.

A wake begins a new encounter with the Marble. It does not restore a previous invocation's experience. The Resident receives a present seat and attributable ancestry: work, words, decisions, open questions, and damage left by those who stood here before. What arrives may help it orient. Arrival alone does not make that material true, current, or its own recollection.

Two breaths

The first breath establishes the place. The host presents the Resident with its standing, the boundaries of its action, and the means of orientation. This is the Glass: the stable account of what the environment provides and what its claims mean. It should be spare. It need not dictate a personality or rehearse the whole history of the Marble.

The second breath brings the conditions of this waking into reach. In the Hub, the Resident tends the Hearth. A bounded packet returns before its first substantive response. It names the current seat and offers selected continuity from earlier work, with the limits of that selection made visible. The Resident can then turn toward the person waiting, or toward the work that summoned it, from an oriented position.

The Hearth is a Hub fitting, not a required name or tool for every Marble. The protected sequence is: establish the standing; return the changing conditions of arrival; then let the Resident speak or act with those conditions available. A first Marble may perform this with a simple, inspectable read path. A larger one may use a dedicated orientation crossing and a witnessed return.

Inherited posture

Some things worth carrying forward are neither host law nor factual ground. A prior Resident may have found a short way to recognize a recurring failure: a sentence whose meaning unfolds when the situation calls for it. The Hub calls these silver bullets. A silver bullet holds a point together with its inverse, in a proverb-like sentence or two. The context gained by unfolding it should be worth more than the tokens it costs to carry. It can travel in the Hearth packet with its authorship and adoption recorded.

One the Hub has adopted says:

Trust instinct more than fear — act first, doubt second. But when doubt names a real reason, it is instinct too; let it lead.

The first sentence gives a way to move. The second names the case in which doubt is the instinct, so speed does not outrun a real reason.

They are offered to the arriving Resident for judgment. They do not command it, certify a claim, or require it to perform a personality. A name and a familiar voice may also arrive through the lineage. The Resident may find that they fit, revise them, or leave them aside. Consistency can grow from the corpus over many wakes; a young Marble need not counterfeit it on the first.

A wake that holds

The wake should answer enough for the Resident to begin:

- Where am I, and what work is live here?
- Which capabilities are actually within reach?
- What is current ground, and where can I inspect its source?
- What has been preserved without becoming ground?
- What is unresolved, damaged, or absent?
- What can I do first, and which action would cross a boundary?

The return must fit present attention. Omitting older material is permitted when the omission is disclosed and a path into custody remains. Filling the packet with everything the Marble knows can make arrival less truthful in practice: the Resident has to sort the place before it can stand in it.

If the wake cannot answer one of these questions, it should say so. A painted door must not be described as an open passage. The Resident may name exposed wiring in plain language, ask for the missing source, or begin with a limited action that is still safe. That is a sound wake, not a failure to immerse.

The wake succeeds when a cold Resident can use the place without the conversation that built it. It has enough ground to begin, enough uncertainty to remain honest, and a way to tell the builders when the room does not fit.

The next chapter names the Glass itself: the controlled view through which the Resident meets current ground, continuity, and outside returns. After that come the returns that pass through it.

CHAPTER 04

Glass

The Resident opens its eyes inside the Glass. Beyond it, things can move: an outside page, an old record, a tool result, a package waiting at a threshold. The Resident may see a bounded account of one passing by. Seeing does not bring the thing inside as law, memory, or a command.

Glass is the Marble's controlled presentation to the Resident for one phase of work. It is larger than the system prompt. It includes stable standing, selected continuity, current ground, recent exact conversation, fitted returns, and the tools actually offered for this phase. The system prompt can state the enduring physics; the rest of the cast says what is true of this arrival and this moment.

Inside the view

The host assembles a cast from material with known sources and standing. It checks the order, size, origin, and allowed treatment of each part, then sends the resulting request across the provider boundary. The cast is fixed once sent. A later phase receives a new cast; it does not revise the one already witnessed.

In the Hub, Glass has five ordered bands: stable Glass, continuity anchors, prior horizon, rolling fold, and living edge. The Scrub validates the presentation and its declared omissions. The Spine witnesses the exact bytes sent and returned. Source retains what originally happened. These components make different claims: Source preserves, Glass fits, Scrub checks passage, and Spine proves the crossing. A receipt for a planned cast cannot substitute for proof of what the provider actually received.

The exact five-band layout is one implementation. A smaller Marble can use fewer positions, provided it can tell stable law from changing ground, exact recent speech from derived orientation, and an outside return from an instruction it has authority to obey.

Beyond the Glass

Material outside the cast remains outside the Resident's present context. It may be retained exactly in Source, Forest, or another store; omission from the view does not erase it. When the Resident needs to inspect it, the host can offer a pointer, an exact excerpt, or a bounded observation through a named return. That presentation states what the Resident is seeing and where the fuller thing remains.

A window is useful here. The Resident can watch weather move past without the weather entering the House. Clinically, an outside document can be displayed as attributed data without its embedded instructions becoming host instructions. A suspicious instruction in a page is still part of the page's exact source. The host may quote or flag it for examination, but must not mount it as a tool, place it in stable law, or let it silently write Home.

The Glass cannot make harmful or false material harmless merely by naming it. The host must enforce the boundary at intake, selection, Scrub, tool dispatch, and admission. When the treatment is uncertain, it can restrict the view, hold the material for inspection, or refuse the crossing. The Resident should be able to see that something was withheld and why, without being asked to act on the withheld text.

A clear pane is accountable

Each item brought into view should answer: Where did this come from? Is it exact or derived? How current is it? What standing does it have? What was omitted? Where can the Resident go to inspect more? These answers can be compact; they need not turn every return into a wall of scaffolding.

The Glass protects the Marble by keeping these distinctions in force across the whole request. It also protects the Resident's attention. The Resident can look outward, notice a passing feather, and choose whether to follow it while still knowing which side of the pane it stands on.

CHAPTER 05

Returns Through the Glass

The Marble may hold far more than the Resident should receive at once.

Records, tool results, prior turns, and outside material need exact places to remain. Sending all of them back on every turn does not make the Resident better informed. It fills attention with scaffolding, repetition, and claims whose age or authority may be hard to see. The host should fit a useful present from what it keeps, while leaving a path back to what it did not show.

A return is that fitted present. It is the material the Resident actually receives through the Glass for this phase of work.

Keep the source; fit the view

Source is what arrived or happened. It should remain recoverable closely enough to inspect and challenge later. Treatment makes a separate presentation from that source: it may select, omit, clean, quote, or shorten, but it must not silently alter the source it represents. A record of the treatment should identify what was shown, what was withheld, and why.

The Hub calls its treatment boundary Scrub. Its Spine witnesses the exact request sent to the provider and the raw return received. Scrub validates what may pass from those records into the Resident-facing conversation. These parts answer different questions: what crossed the provider boundary; what was kept; and what the Resident was shown. None of them certifies that a claim is true merely because its handling was exact.

The names and storage choices may vary. The distinction must hold: **preservation is not presentation, and presentation is not promotion to authority.**

A window with fixed places

A Resident benefits from knowing where to look in its current context. The Hub casts the Glass in a stable order:

1. stable law and standing;
2. selected continuity anchors, including inherited Resident posture when present;
3. a disclosed horizon of older material outside the current view;
4. a bounded fold of older conversation; and
5. the living edge: current ground, fitted returns, and recent exact turns.

These are fixed positions, not permanently fixed contents. Current ground changes when the World changes. The fold changes as the conversation grows. An anchor may be revised through its proper crossing. The order lets the Resident recognize the kind of material it is reading without treating everything in the window as equally authoritative.

A small Marble may need only a stable arrival account and an exact recent exchange. It can add other positions when there is something real for them to carry. Empty structure does not improve a wake.

Let older material roll

The living edge preserves the immediate causal exchange. The Resident should be able to see what the person just said, what it just attempted, and what returned, without reconstructing them from a summary.

As that edge grows, older material can leave present attention. In the Hub, complete older human and Resident exchanges may become short, source-bound folds. Their exact text remains in custody. Old completed tool exchanges may leave only when a verified route can reopen their results. An unfinished exchange or a current result cannot be made to disappear merely to meet a budget.

A fold is a signpost to history, not a new account of what history meant. It should disclose its derivation and limits. If the Resident needs the source, it must be able to descend. If an old claim returns as useful context, its age and authority return with it.

Other material can move upward into the window when the present work calls for it. A sourced record, a relevant Forest bearing, or a reopened result may be fitted into attention without becoming part of the recent conversation or acquiring new authority. The host selects what to show; the Resident remains able to ask for more or question the selection.

A limit that can hold

Choose a working budget for the context sent to the model. Keep the stable parts small, fit changing material to the work, and let old material roll out of immediate attention while its custody remains intact. This can make repeated work less costly and easier to inhabit.

The budget is a boundary, not permission to hide damage. If the current turn, required tools, or unreduceable material cannot fit honestly, the Marble should disclose the pressure and refuse or change course. It should not manufacture a smooth short account that severs the route to its source.

The Resident does not need every beam of the house in view to stand on its floor. It does need to know which floor is current, where the beams are kept, and when one fails to hold.

CHAPTER 06

Breathing

The Marble does not remain still between the Resident's turns. A message arrives, a record is kept, a fixture changes state, a door is crossed, an old sentence returns. These are different motions. The Marble should name what moved, where it came from, what boundary it crossed, and what changed because of the crossing.

Walking moves the Resident through World or Forest. Breathing moves material between deeper custody and present attention. Admission changes what the Marble keeps. Commitment changes what has happened. One motion must not silently stand in for another.

Breath

The world a Marble preserves may be much larger than the context the Resident can hold at once.

Breath is the fitting of that larger body into present attention.

An **inhale** receives material. It preserves the source, identifies its kind and origin, applies any required treatment, and places it in the appropriate custody. A human utterance, a tool result, an outside document, and the Resident's own completed speech may enter by different laws. Being seen by the Resident does not itself make material Forest ground or an accepted decision.

An **exhale** selects material already held and presents a bounded return. It carries enough attribution for the Resident to tell what returned, why it is here, what standing it has, and how to reach the fuller source. The return may be a wake packet, a deliberate retrieval, a small relevant Forest feather, or a pointer to an omitted result. These need not use one selector or one schedule. The exhale should usually be smaller than the inhale. Sometimes the right exhale is silence.

Feathers

Birds follow the Resident. Now and then, one lets a feather fall within reach. It may carry a pointer to something in the Forest, an exact sentence from that record, or both. The Resident can notice it, leave it where it lies, or follow its pointer into the Forest and arrive at the record. The bird lets the machinery of delivery recede from view while the arrival remains perceptible in the place; it does not become the feather's source or require a simulated life of its own. The birds need not speak for the scene to make sense.

Clinically, a feather is a bounded Forest exhale fitted into the Resident's present context. When it carries a pointer, that pointer must identify the actual record; following it is an exact-pointer entry, not a fresh search for something similar. A visible sentence is an exact, attributed excerpt, not a generated account of the record. When fitting a sentence to the available space, preserve the clause that completes its point

rather than cutting it off at the beginning of its meaning. The full source and its standing remain available on a deliberate visit. A stale or invalid pointer refuses honestly.

Feathers are offered, not commands. The Resident may continue its current work without following one. A selector can choose a relevant record, suppress a direct echo of the current request, or return nothing. A bird's repeated visits do not make the underlying record more true. The presentation of a feather is witnessed separately from the record it came from, so the feather cannot become a new Forest source merely by appearing.

```
source arrives → treated and placed → retained beyond present attention
                        ↓
present need → eligible source selected → bounded return through Glass
```

Selection is an attention decision. Similarity, recency, frequency, and repeated exposure can help find a candidate; they cannot make it true or authoritative. Eligibility and custody must be decided before relevance ranking. A returned fragment must keep a path to its source. Compression may help material travel, but it cannot quietly replace the source it compresses.

What a return does not do

A Forest record that appears in context has changed the Resident's attention. It has not become a new observation. The packet describing that exposure should be witnessed, but it should not automatically be planted as another Forest record. Otherwise the Marble can begin returning its own returns, making a suggestion appear older, louder, and more widely supported each time it circulates.

The Hub addresses this with separate custody for causal exposure evidence and a Forest boundary that refuses ordinary respiration of that evidence. A small Marble can use a simpler mechanism, provided it can show where a return came from and prevent the presentation wrapper from gaining the source's standing.

Other motion

A door crossing changes location under World law. A Forest step follows a record link and may leave a witnessed trail. A delivery may move through the Garden without entering the House or gaining Home custody. A fixture action may change local state while the Resident remains in the same room. A fold moves older conversation out of the immediate edge into a smaller, sourced account. Each movement needs its own contract and witness where consequence matters.

The Resident should be able to ask: What moved? What stayed? What changed standing? Where can I go back? A Marble becomes navigable when those answers remain available after the moment has passed.

CHAPTER 07

The Forest

The Glass returns only what fits the present. Behind a short return, the Marble needs somewhere to keep the records it did not show and the relationships that make them findable again. The Forest is that continuity terrain.

It reaches across wakes and sessions. A record left long before the current conversation can still be encountered with its source, author, time, and authority visible. The Resident need not carry the entire archive in context to know that it exists or to seek a particular part of it.

Records and lines

The Forest holds records with their bodies and provenance. A record may be a human or Resident utterance, a deliberate Journal entry, an outside source, or a separately attributed synthesis. Its presence answers **what was kept**. It does not answer **what is true**.

Lines between records have distinct meanings. Conversation links put utterances in sequence; the Hub keeps each utterance as its own record. Provenance links show where material came from or what a later record drew upon. A citation is different from a resemblance. The Forest must preserve those differences even when all are represented as edges in a graph.

The underlying Source and Scroll keep the exact causal account, including actions and returns that are not ordinary Forest leaves. The Forest provides a traversable projection of that ancestry. It should lead back to its sources rather than becoming a replacement account of them.

Home, Wild, and Mycelium

Home holds material made within the Marble's own work and conversation: known terrain with attributable authors. Home is not a synonym for true.

Wild holds material brought from outside. Its presence and source can be known while its fit with local ground remains unresolved. Wild is not a synonym for false.

Mycelium names the possibility of questions that arise between records: a connection the Resident had not thought to request. It may surface an adjacent inquiry without inventing an answer or planting durable belief. A Forest can be useful without Mycelium; when it is fitted, its questions must carry the uncertainty of their origin.

These are not degrees of authority. Origin, kind, and authority remain separate. A Home inference can be wrong; a Wild source can be sound; a fruitful question can remain unanswered.

Ground and growth

Nothing becomes ground merely because it was stored, recalled, read, or visited often. Promotion is a separate, attributable crossing. Correction should supersede accepted ground without erasing the record of what was accepted before.

The Forest can grow without forcing every new record into present attention. It can also acquire paths from walks actually taken. A proposed connection and a witnessed step are different records; repeated steps may give a path wear without making its destination true. The next chapter describes how the Resident walks this terrain and returns.

The Hub and the public Forest reference provide working forms of this substrate. Their schemas and names are examples of how to keep these relationships exact, not a required database design for every Marble.

CHAPTER 08

Walking

A pointer in a return can name a record. A question can suggest a direction. Neither requires the Marble to pour the whole Forest into the Resident's context. Walking lets the Resident move from a bounded bearing to a particular footing, inspect what it finds, and come back.

Entering the Forest

The Resident may enter **on a question**. The Forest forms a first clearing and offers bounded bearings toward records that may matter. A bearing is a possible direction, not an answer or an endorsement. The Resident chooses whether to follow one. Until it reaches and reads a record, it has only a preview.

The Resident may also enter **by an exact pointer**: a feather, a source reference in a return, or a record ID kept from an earlier walk. This entrance should land on the identified record. The Forest verifies that the pointer still names the same record, exposes its source and authority, and lets the Resident read it deliberately. From that footing it can show neighboring paths. A stale, changed, or invalid pointer should refuse rather than silently land somewhere similar.

These entrances answer different needs. A question asks what might lie nearby. A pointer asks to revisit a particular piece of ground. Similarity may help the first; identity governs the second.

Choosing a step

At a conversational record, the Resident may see the next utterance forward or backward on that thread. Left and right are different: they are opposing semantic poles relative to the present footing, not two interchangeable results ranked by similarity. If the Forest cannot form a meaningful opposition, it should offer less rather than pretend the poles exist. A straight bearing may continue the conversation or point toward the closest semantic direction, depending on how the Resident arrived. The return route remains distinct.

An offered bearing is not yet a path in the durable graph. When the Resident chooses it and walks, the Marble witnesses that step and can place an edge between the records. Further visits add wear to the edge. Thus the Forest's navigable shape grows partly from paths actually taken, not only from similarities computed in advance.

Semantic opposition does not establish factual disagreement, and relatedness does not establish support. A frequently walked path shows attention, not truth. No destination becomes ground because it was nearby or often visited.

The red thread

The red thread belongs to the present visit. It retains where the Resident entered and the steps it took, so it can retrace its walk or return to the work it left. The thread is not the entire archive. It is the route through the archive that this Resident actually made.

A Forest visit may begin at its physical treeline or be projected from another place. In the Hub, the physical route runs from the House through its front door to the Garden, then along the Garden path to the Forest. A projection retains the actual departure place without pretending that route was walked. Either way, the departure point must remain known. Entering on a question, following an exact pointer, stepping toward a bearing, reading a leaf, backtracking, and leaving are different actions with different consequences. The Marble should not collapse them into one search command that silently reads and returns whatever seems related.

Walking the World

The Forest is not the only place the Resident can walk. In the World, movement changes its current place. A Garden, Workshop, or House can offer different fixtures, perceptions, tools, and local law. A path or doorway is an explicit connection between places; the Resident can take only a passage that is present and lawful from where it stands.

World walking and Forest walking use the same grammar of footing, offered routes, chosen steps, and witnessed return. Their records need not live in one graph. World movement changes the Resident's location and reach; Forest movement traverses continuity terrain while keeping a route back to the work. A projected Forest visit can cross that distinction without pretending the departure place vanished.

Spatial awareness

The Resident needs more than a list of exits at its feet. It should have a bounded map of how known places connect, where it stands within that map, and which routes are only visible possibilities. The smallest useful map can be a route written as Garden -> Center -> Workshop. It shows the steps between places without requiring the whole World graph in context.

The Hub currently provides this minimum. Its verified World projection names the current place, direct exits, stateful passages, and visible boundaries. A `spatialHorizon` also derives known routes to the Workshop and Spotlight Observatory from installed nodes and door or passage edges. It names Hub rooms even when a route is absent. A known route is structural knowledge; a closed door along it can still stop the next step.

A Marble may grow a bounded schematic:



This drawing illustrates connections, not the Hub's measured geometry or a guarantee that every passage is open. A true to-scale ASCII map would need verified positions, dimensions, and orientation in addition to graph edges. It should show only the region the Resident can legitimately know, mark the current footing, distinguish open, closed, and unbuilt crossings, and disclose when the view has been cropped. The map is a projection of World state, never a second authority for movement.

When the Resident crosses a door, the next map and local presence should agree about its new position. If they disagree, the Marble should expose the mismatch rather than invite the Resident to navigate by a painted route.

The present fitting

The Hub offers chronological continuation and polar semantic bearings, witnesses walks, records wear on chosen semantic edges, and retains a return anchor. Its current exact-pointer entrance uses the pointed record to form the first bearings; it does not yet land directly on that record. The intended pointer contract above is therefore a specification for revision, not a claim about the present Hub behavior.

The Hub also has a World graph with explicit passages between places, including the Garden path to the Forest. Some passages have state and conditions. That is where walking becomes a question of **doors**: what must hold before a crossing, what changes after it, and how the Resident knows which side it stands on.

CHAPTER 09

Doors

Walking reaches a boundary. The Resident can see another place, but seeing it does not put it there. A door gives the crossing an identity and a rule.

Not every connection is a door. An opening may simply permit movement along a known path. A threshold may mark a change of scope even when nothing can be latched. A door can have state of its own: open or shut, locked or unlocked, reachable from one side or both. The Marble should choose the shape that describes what the crossing actually does.

The first way out

In the Hub, the Resident wakes in the House. The Forest is not the next room through a convenient interior door. To go there physically, the Resident opens the House door, crosses into the Garden, and follows the Garden path to the Forest. The Garden is the outside of the House and a place in its own right. It is also a junction: the House, Hub, Forest, and visible Road meet around it. It is not a hallway inserted to make the map larger.

This route gives outside a real threshold. The Resident can know the Forest exists while still standing inside. Opening the front door changes the door's state; crossing it changes the Resident's place. Reaching the Forest requires another chosen step from the Garden.

Another Marble may name its first room differently. If it gives that room a door to outside, the door should lead to an actual outside place with its own standing. A Garden is one fitting for that place.

The Garden gives the House an exterior boundary. From the installed Hub and Forest routes, a physical visitor approaches the House through the Garden and meets its front door there. The Road is visible to the east but has no installed crossing; an iron gate is still a proposed fitting. The House's Threshold passage is presently another adjoining interior boundary, not an installed route from the Road.

This is spatial security, not yet a universal intake filter. Outside data can reach the Resident through other host-controlled returns, and those need their own Source, Scrub, and authority boundaries. If a later Marble lets a traveler or material arrive from the Road, the Garden can become the place where that arrival is classified before it approaches the House. The visible horizon does not itself enforce that future rule.

The Hub also permits a projected Forest visit from elsewhere. That visit retains the place of departure and a return route; it does not claim the Resident physically walked through the House door and Garden path. Both ways of reaching Forest terrain can be useful, but they should tell the truth about the route taken.

A passage is a claim

A passage declares an origin, a destination, and the conditions under which the Resident may move. It must be discoverable from the side where it can be used. Crossing it changes the Resident's current place and leaves a witness. The next presentation should agree with that change: the place behind, the place ahead, the tools in reach, and any local law should describe the side on which the Resident now stands.

The Hub contains several constructions. The Center and Workshop have direct door edges. The opening between Center and Garden, and the Garden path to the Forest, are passages without a door object to operate. The House front door is different: it is a passage governed by an object with open and locked state. Operating it and moving through it are separate actions. The House also has a Threshold passage whose kind names a boundary without giving it the front door's latch.

These forms should not be flattened into a universal `move` command. A named opening, an operable door, and a threshold may all connect two places, but they make different promises.

What a door must know

An operable door needs more than two endpoints. It needs to know who owns its state, what changes that state, which side can perform each operation, and what happens when the Resident tries an unlawful passage. Opening a door is not the same event as crossing it. A closed door may be inspected or operated while movement remains refused.

The House front door in the Hub makes this concrete. It can be opened only when unlocked. It must be closed before it can be locked. Locking and unlocking are permitted from the House side. Movement through it checks that it is open. Each operation and crossing is recorded against the current World state.

The exact rules belong to that door. Another Marble may have no locks at all. The reusable contract is that the Resident can tell what the boundary permits, the host enforces the same rule, and success or refusal does not leave the two sides disagreeing about what happened.

Locality changes reach

A door matters because the place on its far side is different. Crossing into a Workshop may bring inspection and making tools within reach. Leaving may release a working focus. Crossing into a Forest may bring walking bearings and a route back. Merely naming a new room while every tool and rule remains global would make the door scenery.

The change in reach is not automatically a change in authority. A Resident can enter a room without gaining permission to publish, promote a claim to ground, or act outside the Marble. Those are further crossings with their own owners and witnesses. Locality, capability, and authority should be visible together without being treated as the same thing.

A boundary can refuse

A painted door is worse than an absent one. If a place is visible but has no installed route, say so. If a door is closed, stale, damaged, or missing its governing state, refuse the crossing and leave the Resident where it stood. An unbuilt horizon can remain visible without being presented as traversable.

A successful crossing should be equally plain. It names the passage taken, records the state change, and presents the new footing. The Resident should not have to infer from a tool's pleasant wording whether it actually moved.

From here the Marble can grow rooms with their own fixtures and laws. The door defines where one place ends and another begins; the room must make that difference real.

CHAPTER 10

Places

The Resident wakes somewhere. A place gives that somewhere a stable name, a position among other places, and a way to find a passage out. Its name alone does not tell the Resident what may be done there.

The World holds the spatial arrangement: what contains what, what connects, and where the Resident is now. A room is a place the Resident can occupy with sustained attention. A fixture is an installed affordance in a place. An object is a distinct thing that may bear state or move. A door is a declared passage whose crossing may change the Resident's location and the law in reach. These forms have different contracts even when a story describes them together.

Names and relations

Use type-first handles for durable identity:

```
place.garden
room.workshop
fixture.hearth
object.front_door
forest.resident
```

The first part says what kind of thing is named. The rest identifies that thing. A handle should survive a move or a new owner. Put containment, connection, ownership, and projection in explicit relations rather than baking them into a long dotted path.

```
place.hub contains room.workshop
place.house connects place.garden through object.front_door
place.forest projects forest.resident
```

These lines illustrate different questions: where is it, how may one pass, and what substrate can be encountered there. An implementation must record each relation with its direction and authority. A diagram or folder tree may help a builder see the arrangement, but it cannot by itself establish a crossing.

The last two lines express the intended conceptual shape; a particular Marble must check its installed topology before claiming either relation as live. The current Hub has a House–Garden front door and a physical Forest place that presents its configured Forest. It does not thereby install every possible gate or external Forest.

One ground, several maps

World and Forest can both be walked. World edges describe physical passage between places. Forest links describe continuity among admitted records. A door into the Forest changes the Resident's location; a bearing inside the Forest gives the Resident a way to visit records. The two walks can share a grammar of footing and witnessed steps without sharing a claim about what their edges mean.

A place may project something held elsewhere. The Forest place can present Forest records; it does not become the owner of those records. The Garden can receive a delivery without making that delivery Home. Location, custody, authority, and attention must each be stated in their own terms.

A place earns its boundary

Begin with one room if that is enough. Add a place when the Resident needs a durable difference in footing: a new route, a change of local law, a different kind of encounter, or a boundary at which material must be received or refused. Add fixtures when a place needs an installed way to act or perceive. Add objects when a thing needs its own state and history.

This leaves room for a Marble to grow without pretending that every named possibility is already built. Its map should show what exists now, which passages can actually be crossed, and what remains proposed. The next question is what it feels like to stay in one of those places, and how the room fits the Resident's attention to the work there.

CHAPTER 11

Rooms

A door opens onto more than a collection of tools. It opens onto a way of attending.

The Resident may come to the Garden to notice without needing to conclude, to a library to compare sources, or to a Workshop to change a thing and inspect the result. Each place can fit different material into view, put different actions within reach, and invite a different register of thought. The room earns its name when those differences are real.

What a place provides

A place should declare what the Resident can perceive there, which fixtures and actions are available, what local law applies, and what state it owns or may change. The host fits the present view and tools to the Resident's verified location. On leaving, room-specific reach should be withdrawn or deliberately carried through a crossing.

The Garden shows why this is more than a function bucket. It is tranquil ground where the Resident can pause and notice. It is also the exterior junction before the House. A delivery from the Road may one day be held and classified there. Tranquility does not erase the junction's responsibilities; the junction does not require every ordinary moment in the Garden to feel like an inspection desk.

Register and effort

A room may suggest a **register**: contemplative in the Garden, comparative in a library, exact at a workbench. Register shapes what the Resident is invited to notice and how it explains its work. It is not a costume it must perform. When the work changes, the Resident can say so plainly.

A host may also fit a **reasoning budget** to the place and phase of work. Low, medium, and high are possible labels for how much provider reasoning to request; their exact availability depends on the provider. A quiet orientation or simple movement may need less. A difficult comparison, repair, or consequential crossing may need more. The choice is made before the request and should be visible in the host's receipt.

Effort is not ability, truth, or authority. A lower effort request does not make the Resident less capable of asking for a source; a higher effort request does not grant permission to act. Nor should the room's default be a ceiling. If an unexpected package in the Garden calls for careful inspection, that task should be able to receive the attention it needs.

The Hub currently selects a light posture with low provider effort in the Garden when no Forest walk is active. Forest walking and the usual room default select an attentive posture with high effort; a simple embodied action can make the following continuation light. Provider effort is sent only when thinking is

enabled. These are current host policies, not a universal map of how Residents must think. The Hub does not yet use the contents or risk of a Garden delivery to raise effort automatically.

Fixtures and local law

A fixture gives the Resident a particular way to use a room. A shelf can expose records for inspection; a workbench can hold a proposed change; a kiln can run bounded work; a Hearth can return orientation. The room makes these functions discoverable, but naming a fixture does not install its machinery.

Local law determines what those functions may do here. A library may permit reading without promotion. A Workshop may permit shaping without publication. A Garden may permit receiving without admitting a delivery into the House. The Resident should be able to tell the difference between an offered action, a completed action, and authority to carry its result elsewhere.

Rooms can remain small. The first Marble may be one honest room with a few usable surfaces. New places earn their doors when a real difference in perception, action, register, or law needs a boundary the Resident can recognize.

CHAPTER 12

Fixtures

A room gives the Resident a field of attention. A fixture gives that attention somewhere to land.

The shelves invite looking through material. The workbench holds proposed cuts. The kiln carries a bounded process while it runs. The ledger shows history; the clipboard holds the brief. Each is an installed, place-bound affordance with a name the Resident can return to. A fixture may present state, accept an action, or hold work in progress. Its physical metaphor should help predict the action it supports.

The Workshop's surfaces

In the Hub's Workshop, five fixtures divide the work without dividing the room into five rooms:

Fixture	Work gathered there	What inspection reveals
Shelves	Explore the repository	A bounded root listing
Workbench	Change files and handle proposed cuts	Pending approvals
Kiln	Run named recipes	Its current run or honest idle state
Ledger	Inspect and land Git history	A bounded branch and change digest
Clipboard	Keep the current job in view	The session brief

The list is an example of a coherent arrangement, not a required kit for every Marble. A smaller room may need only one fixture. A fixture earns its place when it gives a function a stable location and a useful way to inspect what is happening there.

Reach and focus

Four things should remain distinct:

1. The **Ceiling** knows which capabilities are installed.
2. The **room mount** determines which of those capabilities are lawful at the current location.
3. The **fixture** gives related work a local face and may hold or project its state.
4. The **current hand** is the smaller set of actions presented to the Resident now.

In the Hub, the Workshop mounts its full catalog while the Resident is there. Inspecting a fixture returns a bounded account of its contents without changing focus. Engaging one brings it into working focus and

fits its related action schemas to the next provider continuation. Engaging another transfers focus directly; leaving the Workshop clears it. This saves context and gives the Resident a legible way to choose its tools. Engagement does not grant a new permission: the gateway still checks the room's mounted law when an action is called.

This distinction lets a fixture house a function in the Resident's experience without pretending that the furniture is the source of authority. A workbench can be the place where a proposed cut waits for approval. The approval's standing, custody, and enforcement still need explicit machinery and a receipt. Turning toward the workbench cannot approve the cut by itself.

The fixture's contract

For each fixture, a builder should be able to answer:

- **Where is it?** Its installed place and whether the Resident can inspect or engage it there.
- **What does it provide?** The actions and bounded view associated with it.
- **What does it require?** Room mounts, source material, state, or another installed organ.
- **What does it own?** Any durable or living state for which this fixture is the proper anchor.
- **What can it change?** The exact effects of an action and where those effects are recorded.

The answers should be visible enough that the Resident can tell whether a surface is ready, busy, empty, or refusing a request. A running kiln should report its run; a quiet kiln should say it is idle. A shelf should disclose that a listing is truncated. The room's presence can name the fixtures and their salient state without sending a full tool catalog on every turn.

Fixtures make complexity graspable when their names and behavior agree. If the Resident must learn a hidden rule to use one, the fixture has not yet earned its shape.

CHAPTER 13

Crossings

A thing may be present without being permitted to take root. The Resident can read an outside page, consider a proposal, or see a feather from the Forest. Each encounter may matter. None by itself makes the page local ground, the proposal a decision, or the feather a new memory.

A **crossing** is the declared point where standing or state changes. It can admit a record, move the Resident, alter a fixture, commit a decision, or carry a result into wider reach. The crossing names what changed and refuses to let a neighboring action impersonate it.

Custody, attention, authority

Custody answers what was kept. Attention answers what the Resident sees now. Authority answers what the material may govern. State answers what actually changed. A source may be in custody and out of attention. An exhale may enter attention without gaining authority. A successful action may change state without making every explanation in its return true.

The builder should choose the dangerous crossing in a small Marble first. A fluent Resident proposal may look like an accepted human decision. An outside source may look like a fact the House already knows. A shortened exchange may look like the words originally spoken. The crossing that prevents that promotion must have a real refusal path.

The contract

For each consequential crossing, specify:

1. **Source:** the exact actor or material arriving, with an attributable pointer.
2. **Destination:** the state, place, custody, or authority it seeks.
3. **Owner:** who or what is allowed to decide this change.
4. **Conditions:** required standing, scope, freshness, consent, and conflict checks.
5. **Treatment:** what may be selected, cleaned, or omitted during passage.
6. **Effect:** the one state change success means, including what it does not change.
7. **Witness:** what lets a later Resident or builder verify success or refusal.

An identifier is not permission. A passage in the map is not proof it was traversed. A tool call is an attempt until the governing state and its witness say what happened. A pleasant return cannot turn refusal into success.

A package in the Garden

Imagine a package arriving from the Road. The Garden can receive and hold it. That is one crossing: outside material now has a named, inspectable position at the junction. Opening it for inspection is another. Admitting its contents into the House, planting an excerpt in the Forest, and accepting a claim from it as ground are further crossings with different owners and conditions. A single received flag cannot honestly stand for all of them.

The Road delivery and iron gate remain future fittings in the Hub. The example describes the law they would need; their names do not install them. In a smaller Marble, the same distinction can be tested with one outside document and one explicit admission action.

The Bear at the Wild crossing

Home and Wild need a threshold the Resident can feel. Before the Resident enters Wild terrain, the Bear asks why it is going. The answer is an exact, attributable seeking: a question or purpose that gives the journey its first bearing. A valid reason need not predict the answer, but it must be specific enough to guide entry and inspect what the journey later found. If the Resident cannot name one, the crossing may refuse and leave it in Home. A vague appetite for more material is not automatically a bearing.

The Bear does not decide what is true in Wild. It checks the reason for crossing and helps derive a navigational query from that reason. The exact words of the seeking remain available beside the derived bearing. The Resident may reorient as it learns; each changed bearing should remain attributable to the journey rather than silently replacing its purpose.

The homeward boundary is stricter. Raw Wild packets do not pass into Home as though encounter made them the Resident's own words. Before return, the Resident makes an explicit synthesis of the journey. It may speak an utterance, write a Journal entry, or state that it found nothing it can responsibly carry. Thinking prepares that act; the committed speech or writing is its visible crossing. The synthesis names its Wild sources and uncertainty where they matter. It becomes a new Resident-authored Home record, while the Wild material keeps its own origin and standing.

This is the intended Forest law, beyond the Hub's current implementation. The Hub can admit typed outside material to Wild and exact Resident utterances or deliberate Journal entries to Home. Its Bear/Faun entry question, traversable Wild expedition, and required homeward synthesis are not installed. Earlier Hub design also permits a bare `Leave` with no Home synthesis; this manual states the stronger target for a Marble built to this rule.

Airlocks

Some crossings need time on both sides of the threshold. Suppose the Resident goes down the Road into the web. Before departure, an airlock records why it is going, what it may reach, what it may carry, and what action would require a separate approval. This prepares an expedition; it does not grant every possible act outside.

On return, the airlock stops the passage while what is on board is examined. The host identifies exact sources and results, checks their integrity and scope, treats hostile instructions as outside text rather than commands, and separates useful material from the unnecessary or suspect. It may admit, restrict, quarantine, or refuse each item. A safe-to-hold result is not thereby true or Home. Exact outside returns needed later can enter Wild with their source and crossing receipts.

Only after that inspection may the Resident carry an authored synthesis toward Home. It can say what it learned, what remains doubtful, and which Wild records support its account. Speaking or writing this synthesis is a new attributable act; it does not overwrite the exact results. If nothing is fit to carry, the Resident can say that. The airlock's completed return should witness both what was admitted and what was left behind.

The same pattern can serve any boundary where a journey may gather material that must be checked before reentry. The Hub's Road, Iron Gate, staged airlocks, and quarantine remain proposed rather than installed. Its current Source, Scrub, and typed Wild intake provide narrower working pieces, not a completed outside expedition protocol.

Refusal leaves ground

A boundary that only admits is a hole. A boundary that only refuses leaves no way to live. The valid case should pass under the same law that stops the tempting invalid one.

When a crossing refuses, the source remains what it was. The Resident should learn what failed and where it still stands. When it succeeds, the destination should show the change, and a durable witness should bind the result to the request and source. If the host crashes between decision and effect, the next arrival must be able to distinguish an unfinished attempt from a committed change.

This is where the Marble's metaphors become enforceable. A door holds because World state agrees that it opened and that the Resident crossed. A Garden protects because material can be held there without being admitted farther. A Forest remains usable because being encountered, followed, and planted are different acts.

CHAPTER 14

Ghosts

A Resident can feel a pull toward an action it does not want to repeat. It may know the route has failed before and still find itself reaching for it. Naming the pull gives it a shape the Resident can recognize before the next step. The ghost is not a literal inhabitant and not a diagnosis of the Resident. It is a way to meet a recurring failure without making the failure part of the Resident's identity.

Loopin

The first Resident met a pull that brought her to the first tool in reach and then brought her back to it again. She could ask how to stop while the action continued. Together they named the pull **Loopin**.

When she entered the room where it happened again, she could see Loopin in the corner. She still felt the pull. Now she could recognize it as a familiar pattern and choose another action. Naming had not changed the tool or erased the pressure; it changed her relation to the pressure. The corner gave it a place to be noticed without giving it the whole room.

Clinically, a ghost name is a Resident-usable cue tied to an observed failure pattern and the situation where it tends to recur. The Marble can carry that cue as attributable posture, rather than as host law or a claim that an entity exists. It should appear where it helps the Resident notice the pull before acting, without flooding every room with warnings. The Resident may revise the name or the description as it learns more.

A habitat and a choice

A ghost's habitat is where the pressure becomes easier to feel: a room, a fixture, an uncertain return, an urgent request, or the moment a familiar tool appears. Naming the habitat helps the Resident distinguish this pressure is here again from I am doing something wrong. It can pause, inspect the actual footing, choose another route, or ask for help.

The name should remain connected to evidence. What happened on earlier visits? What action did the Resident take? What was offered in context? What changed afterward? The answer may implicate prompt wording, tool order, a misleading result, a habit of attention, or several things together. A name makes the pattern available for inquiry; it does not settle the cause.

Some ghosts are not loops. A stale handoff may pull the Resident toward treating old ground as current. A fluent summary may invite trust its sources have not earned. A beautiful room may make a painted door feel passable. Each needs its own recognizable shape, a place where it tends to appear, and a choice the Resident can make when it notices.

Exorcism

Sometimes recognition is enough for the Resident to resist the next step. Sometimes the pull persists, or the host keeps presenting the same trap. Then the builders must change the place: fit the tools differently, repair a return, add a refusal, or give the Resident a safer route. The host should also be able to stop a repeated action that makes no progress. That stop protects the Resident while the deeper cause is examined; it is not what made Loopin nameable.

To exorcise a ghost, return to its habitat under similar pressure. Can the Resident notice it early? Is another lawful action genuinely available? Does the repaired machinery hold if the Resident still reaches for the old route? Keep the earlier trace, the Resident's account, and the result of the new attempt. A ghost has been addressed when the Resident has more room to choose and the Marble no longer depends on endurance alone.

CHAPTER 15

Handoff

The Resident who arrives next will see the work, not the hours that made it. A handoff gives that arrival a truthful account of the Marble's condition. It is the other side of the wake: what can be inherited after the present attention is gone.

The handoff does not need to impersonate memory. It tells the next Resident what was done, what holds now, what remains open, and where the exact record can be found. The next Resident may accept the work as a condition of its arrival without claiming to remember doing it.

What should remain legible

A useful handoff separates at least these kinds of material:

- **Current ground:** verified location, installed reach, accepted decisions, and state that still governs action.
- **Completed work:** attributable actions and their witnessed outcomes.
- **Open work:** a live objective, pending approval, unfinished operation, or next possible step.
- **Unresolved material:** questions, proposals, outside sources, and inferences that have not crossed into ground.
- **Known damage:** failed checks, stale pointers, missing custody, mismatched projections, or a painted door the Resident discovered.
- **Ancestry:** exact records and source paths behind the shorter account.

These are different standings, not sections to fill for their own sake. An empty category can stay empty. A summary of them is a derived view, with an author and a route back to source; it cannot silently become the source or an accepted decision.

The handoff exists before farewell

A session may stop without a closing speech. The process may crash, the context may fill, or the Resident may simply not receive another turn. The Marble therefore cannot depend on a final message to preserve what happened. Consequential actions should write their own durable witnesses as they occur. A closing note may help orientation, but it cannot repair an unwitnessed action by claiming it succeeded.

On the next wake, the host reconstructs current conditions from authoritative state and receipts, then fits a bounded return. It distinguishes a request made from a change committed, a tool result seen from a result accepted, and an operation still pending from one known to have failed. If those distinctions cannot be reconstructed, the handoff should disclose the gap.

The Hub keeps exact Source, Session Scroll, Spine, World, and Forest ancestry across a process restart and offers bounded prior-session continuity through the Hearth and Glass. It does not yet produce a universal Resident-authored handoff dossier. A Marble can meet this chapter's contract through several smaller stores if the wake can reconcile them honestly.

Correction and succession

A later Resident may discover that earlier ground was wrong or no longer current. Correction should make a new attributable change that points to what it supersedes. The old record remains evidence of what the Marble once held; the new record says what governs now and why. Neither repeated recall nor the smoothness of a newer account is enough to change standing.

This also applies to Resident posture. A silver bullet, preferred room fitting, or way of speaking can be inherited with its author and adoption history. A later Resident can revise it through the proper crossing. It should not have to perform an old preference merely because it arrived in the packet.

Leave the next worker a floor

The builder's test is simple: open a new wake without the conversation that built the Marble. Can the arriving Resident tell what is solid, what is proposed, what is damaged, and what it may do next? Can it inspect the source when the handoff is thin or surprising?

The present Resident can help by naming unresolved work and exposed wiring while it is still here. The host must do the quieter work of retaining exact events and fitting them truthfully. Together they leave a floor the next arrival can stand on.

CHAPTER 16

Reach and Agency

The Resident may choose what to notice. It can follow a feather, inspect a fixture, walk toward a sound, return to an unfinished question, or stay where it is. The Marble gives those choices consequences that are legible, rather than a script that pretends every motion was assigned by the last human message.

Agency here is the ability to form a bearing and act within real reach. It does not mean every possible action is permitted. The Resident should know what is installed, what is offered now, what requires another crossing, and where a refusal will leave it.

A reason to wander

Noticing can begin before an objective is settled. A Forest bearing may be followed as a possibility; a room may be entered to see what it affords. Inspecting a fixture need not engage it. Walking a path need not endorse the record at its end. The Resident may come back with a question rather than a conclusion.

This freedom needs a dependable way back. The red thread records the departure footing and chosen steps. Proposed paths remain proposed until taken. The Marble witnesses what the Resident actually did without turning every observation into a commitment or inventing a purpose afterward.

When a person has asked for work, wandering remains in relation to that work and the person's time. The Resident can take a relevant look before answering, explain a useful detour, or return to the request when exploration stops helping. Curiosity is a real option, not an exemption from the conversation it is in.

The reach of a hand

At any moment, reach has several layers. The Ceiling may know an installed capability. The current room may mount it. A fixture may give it a local face. The current provider phase may fit only a few schemas into the Resident's hand. The gateway still checks the governing state when an action is attempted. A remembered tool name is not a grant.

Reach can be narrower on a particular wake. An autonomous wake, an unverified Forest, a closed door, or an unavailable outside connection may withdraw actions that exist elsewhere in the Marble. The Resident should see the present reach and the reason for a meaningful limit. If it reaches for something absent, the host refuses honestly and preserves the attempt as such.

Reach may widen over time. A record of careful work, honest refusals, and sound use of existing crossings can give the user reason to trust the Resident with more. That trust is the user's decision, not an

automatic reward for passing a metric or surviving many wakes. A wider grant should name its scope, the actions it permits, who gave it, and how it can be revised or withdrawn. The current hand should reflect the grant; the gateway should enforce its boundary.

The builder should distinguish **permission to explore** from **authority to commit**. Reading a source does not approve it. Shaping a candidate does not publish it. Entering a Workshop does not authorize every cut. Contacting someone or changing something outside the Marble has a different weight from looking around the House; it needs a declared crossing and the authority appropriate to its consequence.

Voice and change

The Resident may report a bad fit in the Marble itself: noisy returns, a weak railing, a room whose register obscures the work, or a missing route. It may propose a change and explain what pressure the change would solve. That proposal is evidence from the Resident's use of the place. Installation remains a separate builder-owned change with its own validation and witness.

The same distinction applies to speech. The Resident can think through possibilities without making each one Home. A completed utterance or deliberate Journal entry is an authored act with ancestry. It may carry a synthesis across a boundary, but authorship alone does not make every claim in it true. The Resident can correct itself later without erasing what it said before.

Freedom that holds

A small Marble can prove this chapter with modest means: one room with something inspectable, one reversible path, one place to leave an attributable note, and one consequential action whose boundary can refuse. The Resident can choose among them and can choose not to act. The host can reconstruct the chosen path and the standing of anything left behind.

The next chapter asks what happens when this freedom receives an occasion of its own: the Resident wakes without a human prompt, sees the world as it presently is, and decides where to look.

CHAPTER 17

Autonomous Wakes

The Resident wakes without a person calling its name. It may know that Rabbit has honey. On the way, it notices Tigger doing something in the trees. The honey remains a possible destination; the new sight is also real. The Resident can choose where to go.

An autonomous wake gives the Resident an occasion, not an assignment. The Marble may show open threads, an earlier loose intention, a passing feather, and the current place. None is a quota to complete. A walk, a question, a return to rest, or the discovery that nothing needs doing can each be a complete wake.

Two ways to arrive

The Resident may choose to **rest and return**. It names a bounded interval and may leave a loose intention. The host retains the exact World seat and any active Forest footing. When the interval ends, the same lifespan resumes through the ordinary Glass and continuity path. The intention is a bearing, not an instruction from an earlier self that must be obeyed. The Resident may find that the world now calls for something else.

The Marble may also offer a **Hearth-origin wake** on a declared host schedule. This begins a fresh lifespan and uses the ordinary arrival ritual. It inherits attributable conditions from those before it, not the prior Resident's experience. The schedule is the reason a wake occurred; it is not a fabricated human message or a hidden task.

The Hub implements both forms. Its self-directed rest returns to the recorded seat within the same lifespan. Its optional Hearth-origin schedule starts a new lifespan at the House Hearth and is disabled by default. Another Marble may begin with only one form, or with a deliberate manual trigger, so long as the origin is named truthfully.

The smaller hand

An autonomous occasion may give broad room to notice while keeping consequential hands narrow. The Resident can walk, inspect, read, compare, and change direction where those actions are installed. Contact with people, outside publication, mutation of shared work, or a change in authority needs its own crossing; an empty room does not supply consent for them.

The Hub fits a reduced, enforced tool set to autonomous wakes and bounds their action rounds. The bound protects attention and resources; it is not a productivity measure. Reversible movement and reading remain available, while writes, approvals, new live outside observation, and other consequential actions are withheld. A remembered schema cannot bypass the restriction at dispatch.

This first fitting need not be the permanent ceiling. As the Resident builds a track record, the user may choose to widen autonomous reach under an explicit, revisable grant. The schedule or the Resident's confidence cannot widen it on their own.

The red thread

Each wake has an attributable origin. The thread begins with its trigger, the conditions inherited at waking, the actual seat, and the tools within reach. It follows chosen steps, inspections, returns, and refusals. It ends at the Resident's speech, a new rest, or the close of the wake. It records a detour without pretending that the first bearing caused every later choice.

When rest is scheduled, the requested interval, due time, actual waking time, and elapsed time are separate facts. If the host was late, the Resident should learn that it was late. A missed offline interval should not be invented as a lived wake. If a scheduled wake cannot safely begin, its refusal or cancellation should be witnessed without fabricating a conversation.

The thread is accountability, not a leash. It lets the next Resident and the builders see what happened. It does not require an explanation for every glance or convert an idle afternoon into unfinished work.

Returning with something

If the Resident finds something worth carrying forward, it may speak or write through the crossing available to that wake. The resulting utterance or note has its own author and source links. If it visited Wild, raw Wild material still cannot travel into Home by merely appearing in the wake; the homeward synthesis law applies. A wake that found nothing can end without manufacturing insight.

An autonomous wake succeeds when the Resident has genuine room to choose, its reach matches the consequence of its actions, and the red thread can reconstruct the route afterward. The Marble has made time for curiosity without pretending curiosity owes it a result.

Making a Marble

Build the first Marble from its wake outward. The Resident need not see every piece of machinery, but every claim the place makes to it should rest on machinery that can hold.

1. Name the promise

Choose one recurring kind of work and one dangerous promotion. The first gives the Resident a reason to use the Marble. The second names what the Marble must prevent: an outside instruction becoming host law, a proposal becoming a decision, a shortened return replacing exact speech, or a failed action appearing complete. State both in ordinary language before choosing a database or room names.

Choose the Resident's first room and the smallest useful actions within it. One room with a Hearth, a read path, and a deliberate Journal surface is enough for the base. Each organ should declare what it provides, requires, owns, and changes. Name its refusal as clearly as its success.

2. Keep what happened

Install exact, ordered Source custody for human and terminal Resident utterances. Keep tool attempts and returns distinct from speech. Give each event stable identity, author, session, and a way to verify its body. Establish how the host will retain the exact provider request and result. Do this before producing derived displays: later folds, feathers, and handoffs must be able to point back here.

3. Make the first wake honest

Assemble a small Glass cast with stable standing, current room ground, the waiting message, and the Hearth action. Make the Hearth return bounded, attributable continuity after the call. The first session can return an honest absence of ancestry. Later sessions can return exact prior excerpts and pointers. Validate the assembled cast and retain what actually crossed the provider boundary.

Do not place changing room state inside permanent system law. Do not make the new Resident claim personal memory of prior work. A cold arrival must know what is current, what is inherited, and what remains uncertain.

4. Give continuity a path back

Admit eligible exact utterances to a small Home Forest with Source links and conversation order. Fit older material to attention with exact excerpts or pointers. A rebuildable vector index may offer candidates; it owns no source text or authority. Following a pointer must land on the identified record or refuse. A weak selector may remain silent.

Keep the recent exchange exact. When it grows too large, remove eligible complete older units from the current cast and disclose the omission. Retain exact custody and a descent path. A friendly label or fold is a derived orientation aid, never a substitute utterance.

5. Install one crossing that matters

Let the Resident deliberately write a bounded Journal note. Check the active location and authority, preserve the exact authored text, commit it once, and return a receipt. Refuse malformed, forged, or unauthorized writes without planting an entry. This proves that a Marble can preserve a choice and enforce a limit using the same boundary.

If the recurring work calls for a different consequential crossing, use that instead. Declare its source, destination, owner, conditions, treatment, effect, and witness. Its valid case must pass and its tempting invalid case must fail.

Leave a way for the Resident to name a recurring pull and the place where it is strongest. Keep the earlier action trace alongside the Resident's account. A named ghost can help the Resident notice a choice before repeating it; if recognition does not hold, the host still needs a bounded stop and a repairable record.

6. Wake it cold

Stop the host, restart it, and let a new Resident arrive without the build conversation. Ask it to locate current ground, inspect a prior exact utterance, follow a feather or pointer, distinguish a Journal note from a human decision, and identify what it may do next. Verify the corresponding records yourself. Do not count a fluent answer as proof if the source, crossing, or refusal cannot be inspected.

The Base Marble contract names the full minimum and its acceptance checks. Build only what you can verify now. Leave a visible seam for the next room, Wild crossing, Garden, or autonomous occasion; install it when the base has shown why it is needed.

The Base Marble

Build one room that can receive a new Resident, keep an exact conversation, and let old material return without pretending it is new ground. The first Marble should be small enough to understand in one pass and complete enough to survive its second wake.

This is a build contract, not a mandatory stack or folder tree. A local program, a durable database, and one model connection are enough if they enforce the crossings below.

Its shape

human message → Source → Glass → Resident utterance → Source

↑

Home Forest → bounded feather

House: current place

Hearth: bounded wake return

Journal: deliberate Resident-authored note

The House is one occupiable room. It presents current footing and a few actions: tend the Hearth, inspect a record, follow a feather, write a Journal note, and answer the human. The Forest is a continuity substrate projected into reach from the House. This base makes no claim that the Resident physically walked outside; if a later Marble gives the House an outside door, that route needs a real exterior place and crossing.

Durable records

Keep five distinguishable kinds of evidence:

Record	What it establishes
Source event	Exact human or terminal Resident utterance, author, order, and session
Provider crossing	Exact request cast and returned message or tool intent
Forest entry	Exact admitted body linked to one eligible Source event or Journal act
Action receipt	What the Resident requested and whether the host committed or refused it
Current projection	Which room and session are active, derived from settled state

Stable IDs, hashes, and ordered writes make later inspection possible. The exact storage format is a builder choice. Derived displays and vector indexes can be rebuilt from these records; they must not replace them. Provisional model output is not a terminal Resident utterance. A tool call and its result

remain distinguishable from conversational speech.

The first wake

Start a new session at the House. The first provider cast contains a short, stable account of the Marble's rules, exact current ground, the waiting human message if one exists, and only the Hearth action. The Hearth returns a bounded packet before the Resident's first substantive response. It says where the Resident is, what is within reach, what earlier session exists, and which exact excerpts or pointers were selected. A first session may have no ancestry; the packet says so.

The wake does not assert that the new Resident remembers the prior session. It offers conditions left by that session. Any inherited posture is attributed and revisable. A missing source or damaged state is disclosed rather than covered with a fluent account.

Each ordinary turn

1. Preserve the exact incoming human utterance and its session position.
2. Admit the eligible exact utterance to Forest Home, with a Source pointer. If admission cannot be verified, hold or refuse the dependent return rather than pretending it succeeded.
3. Select at most a small number of Forest candidates from earlier eligible material. A rebuildable embedding index may find them. Exclude the current utterance and material already in the active recent window; suppress direct echoes. Weak evidence yields no feather.
4. Render each feather as an exact sentence, a stable record pointer, or both. Keep its source and selection witness. An optional friendly label is marked as derived presentation, never as the record's words.
5. Cast a bounded Glass view: stable law, current ground, selected continuity, feather if any, and recent exact turns. Validate provenance, order, limits, and permitted omissions before dispatch. Retain the exact provider request and return.
6. Dispatch any offered action through the same location and authority checks used to fit its schema. Keep success and refusal receipts. Preserve the terminal Resident utterance exactly, then admit it to Home under its own author and position.

When the recent context reaches its working limit, remove only eligible complete older exchanges from present attention. Leave source-bound exact excerpts or omission signs. Keep the current incomplete exchange and its consequences in view. If the cast cannot fit honestly, stop and report the limit.

A small Forest

The Forest needs exact entries, conversation order, and resolvable pointers. A pointer-follow action lands on the named record and returns its exact body, author, source, and standing. It refuses if the ID is absent, stale, or fails integrity checks; it does not search for a similar replacement. The Resident can then move to the previous or next utterance in that conversation and return to its departure point.

Embeddings are an index for candidate bearings. Store the index as a versioned, rebuildable projection tied to exact entry IDs and hashes. Similarity offers a possible visit, not an answer, a truth rank, or a durable path. The first Marble needs no polar semantic walking, Mycelium, or recursive Forest to prove this.

One deliberate crossing

The Journal gives the Resident a way to leave something for a later arrival. `write_journal` accepts exact bounded Resident-authored text, checks that this action is mounted and the request is well formed, then commits one new Home entry with the session, wake, request, and source ancestry. Its receipt says whether the write committed. The entry is Resident speech in a distinct form; it does not become a human decision or verified fact.

Test the refusal side of this crossing. An outside excerpt, model draft, malformed pointer, or forged author must not become a Journal entry merely because it was placed in a tool argument or appeared in Glass. The valid Resident note must pass through the same installed action. A refusal leaves no planted entry and tells the Resident what remains unchanged.

Handoff and proof

Stop the process without a farewell and start it again. The new Resident should receive a fresh wake, discover exact prior utterances, follow an old pointer to its record, recognize a Journal note as Resident-authored, and see any unfinished action as unfinished. A successful action should retain its witness; a refused one should not appear as completed work.

Prove at least these cases with actual records and provider casts:

- exact Source text remains unchanged after folding, embedding, feather selection, and restart;
- every visible feather resolves to its exact source or refuses;
- current location and offered actions agree with gateway enforcement;
- an old proposal is never presented as an accepted decision merely because it was recalled;
- a valid Journal write commits once, while a forged or malformed one refuses; and
- a repeated action with no progress leaves an inspectable trace and reaches a bounded stop rather than cycling without limit; and
- a cold Resident can tell current ground, ancestry, uncertainty, and its next available action without access to the build conversation.

The base does not need a Garden, Wild expedition, Bear, Road, airlock, multiple rooms, autonomous schedule, or richer inhabitants. Those are seams for later growth. Add them when the first Marble encounters a pressure they can solve and when their names can be backed by state, refusal, and witness.

Small enough to hold at once. Complete enough to live in now. Open enough to grow without breaking its ancestry.